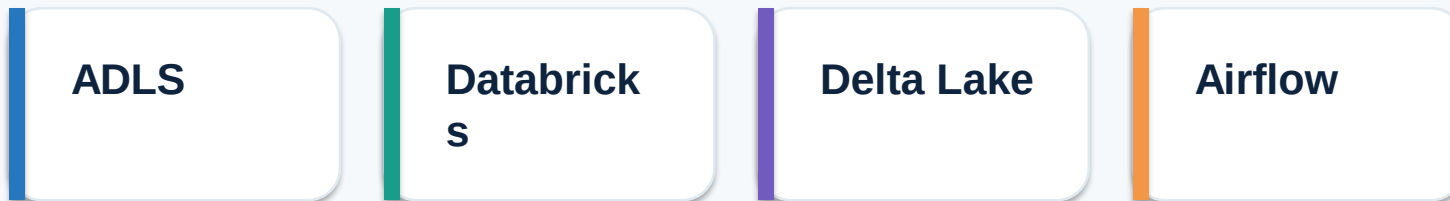
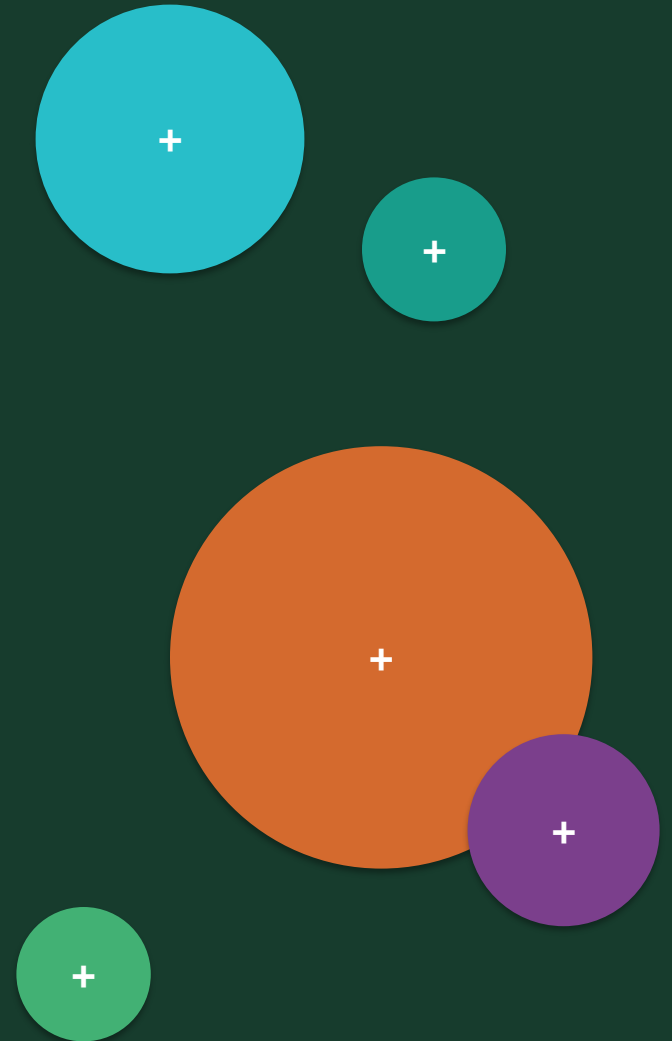


Grocery Sales Pipeline

From raw retail data to trusted sales, inventory and customer insights.



Daily batch • Data Quality • Medallion Architecture • Alerts



Project Overview

SOURCE

Kaggle Heart Disease Dataset / transaction health records delivered as CSV files.

Raw path pattern:
abfss://bucket/raw/grocery_records/YYYY
Y/MM/DD/

PROCESSING

Databricks + PySpark process grocery data through Bronze, Silver and Gold Delta layers.

CONSUMPTION

Gold datasets power category, sales and inventory analytics with dashboards and alerts.

The pipeline is designed for reliable daily processing with validation, error handling and orchestration.

01

INGEST

Capture retail data

02

CLEAN

Standardize & validate

03

ANALYZE

Calculate sales metrics

04

MONITOR

Alert & troubleshoot

What the solution achieves

01

Reliable ingestion

Bring retail transactions and reference files into ADLS with controlled ingestion.

02

Medallion design

Implement Bronze → Silver → Gold for progressively trusted data.

03

Data quality

Handle missing values, duplicates, invalid records and schema issues.

04

Sales Performance analytics

Create sales, inventory and customer performance metrics.

05

Monitoring

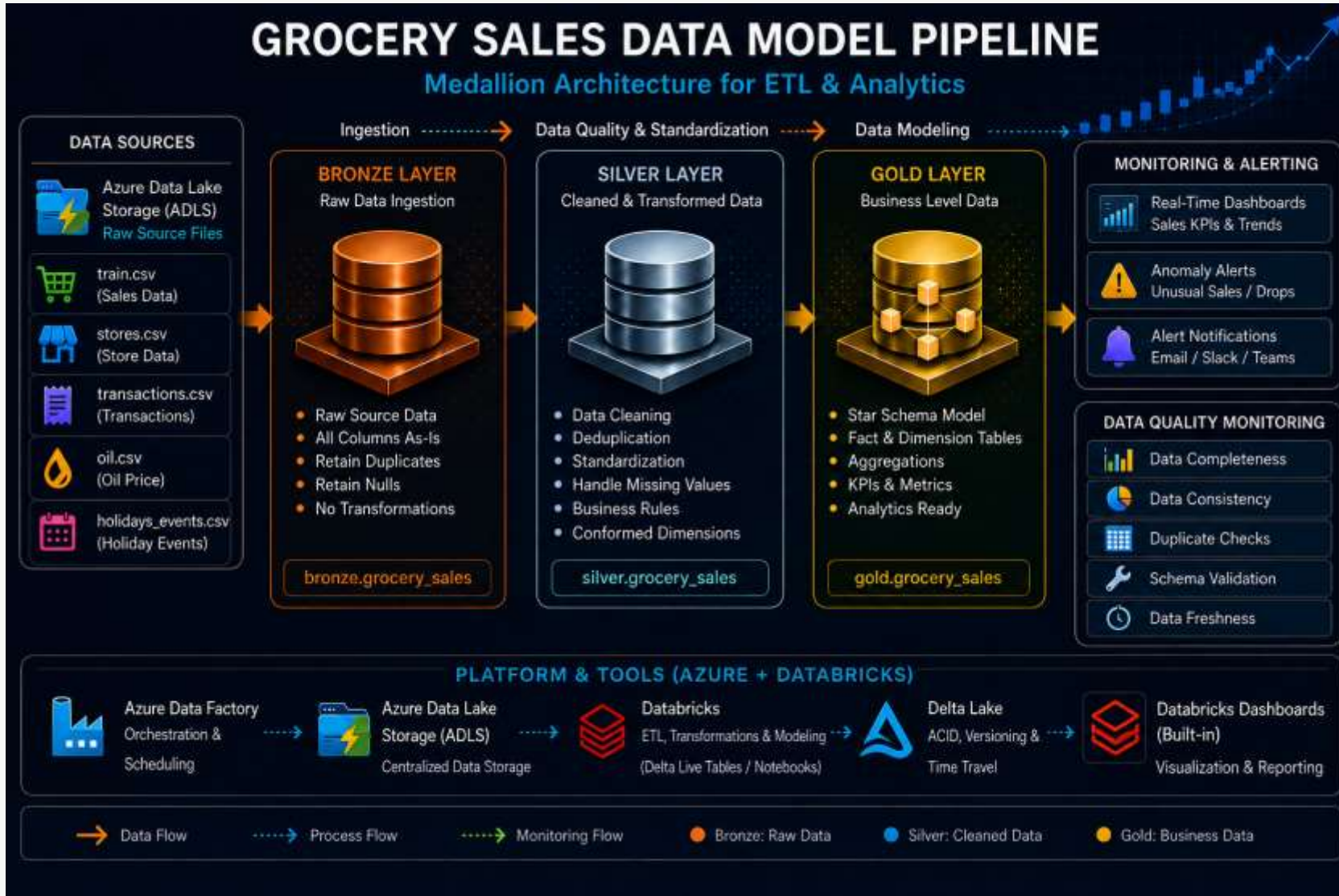
Use Slack alerts and logs for failures and anomalies.

06

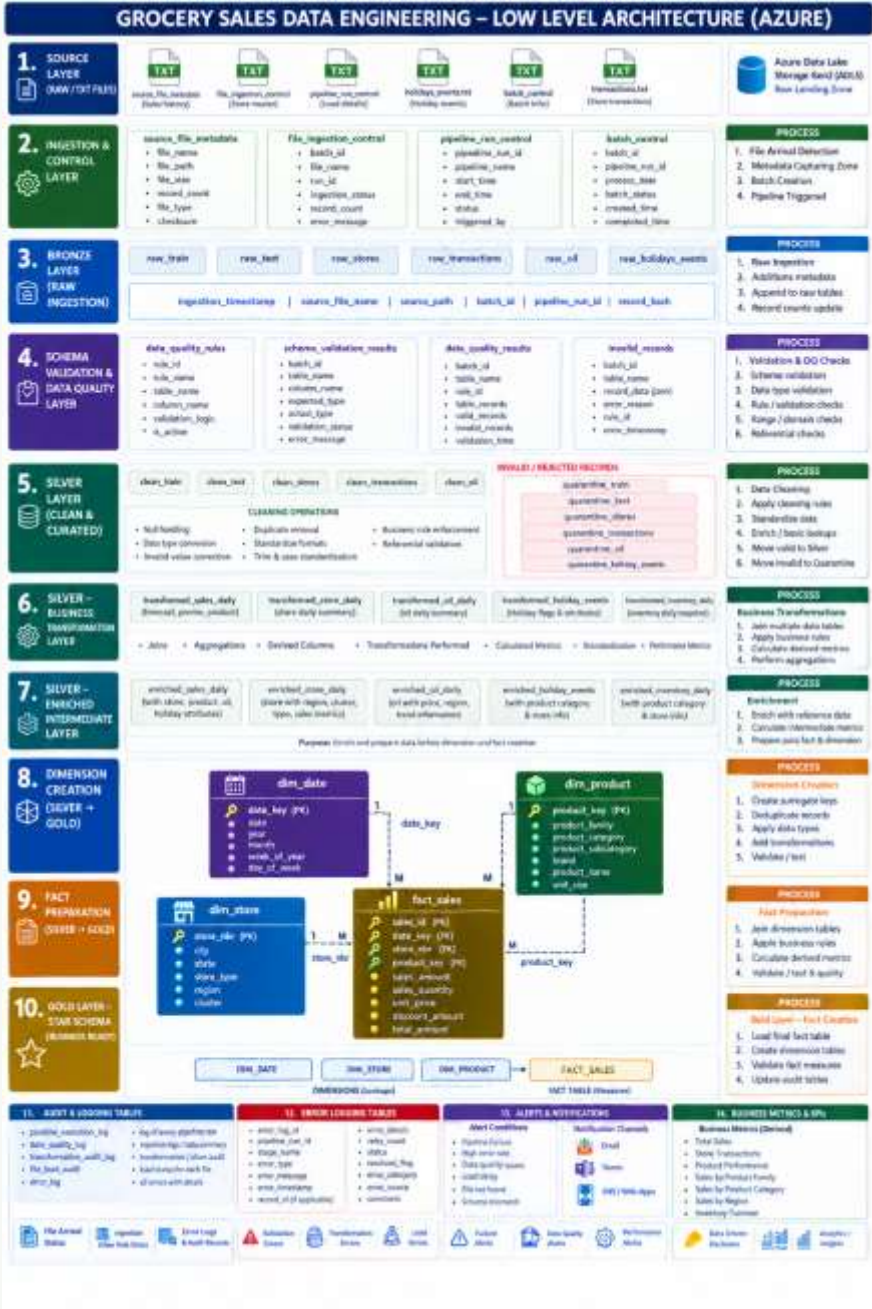
Automation

Orchestrate and validate the pipeline with Airflow and PyTest.

Grocery Sales Data Model Pipeline



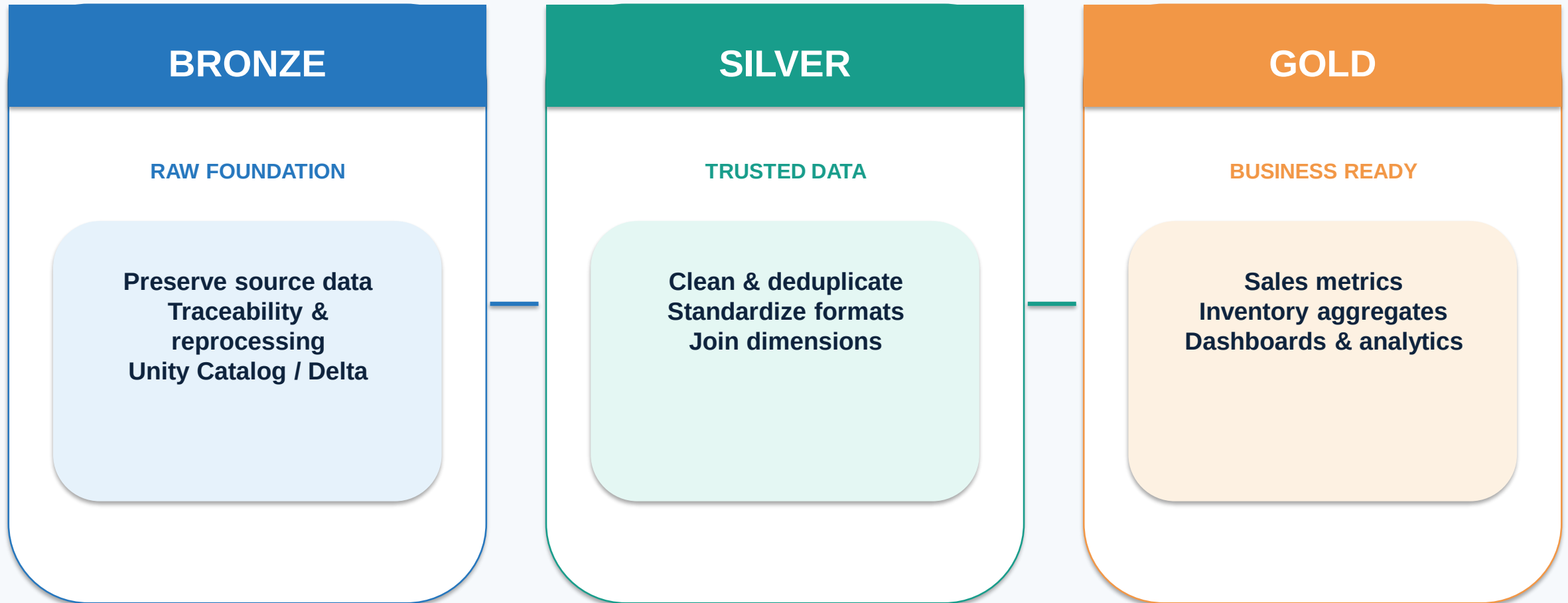
05 • LOW-LEVEL GROCERY SALES DATA ENGINEERING ARCHITECTURE



Storage & Target Design

RAW	Original CSV	abfss://raw@grocerylake.dfs.core.windows.net/YYYY/MM/DD/
PARQUET	Ingested format	Columnar files for scalable processing
BRONZE	grocery_catalog.raw.transactions	Raw Delta table
SILVER	grocery_catalog.silver.clean_transactions	Cleaned & standardized
GOLD	grocery_catalog.gold.sales_metrics	Aggregated sales and inventory metrics
LOGS	grocery_catalog.logs.etl_errors	Errors & monitoring details

Bronze → Silver → Gold



Bronze Layer — Raw Data Preservation

TARGET TABLE

grocery_catalog.raw.transaction_records

Delta table containing raw transaction records after ingestion.

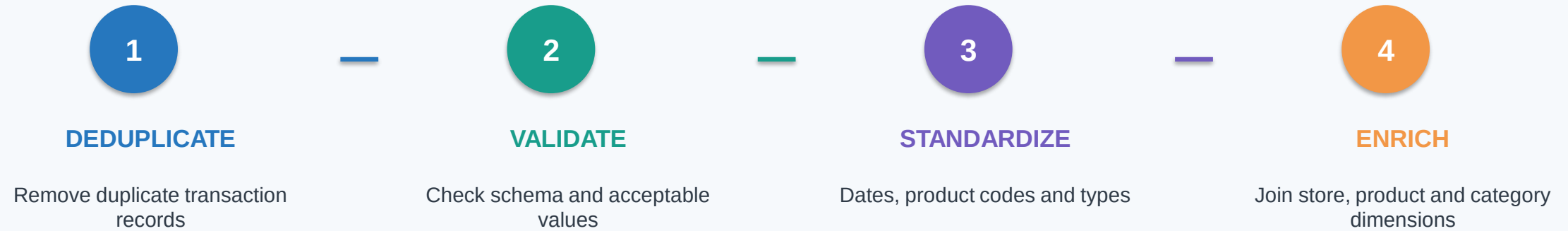
IMPLEMENTATION

- Unity Catalog governance
- External location where applicable
- Preserve source structure
- Capture ingestion metadata
- Support reprocessing

KEY PRINCIPLE

Bronze preserves raw retail data for traceability and downstream processing.

Silver Layer — Cleansing & Transformation



OUTPUT

grocery_catalog.silver.clean_transactions

Source Dataset & Data Quality

SOURCE DATASET

- Sales, stores, products and transaction CSV files
- Product, category, price and quantity attributes
- Daily cloud landing pattern
- Source metadata captured for traceability
- Raw data preserved for reprocessing

QUALITY CHECKS



Missing / NULL values

Detect and handle incomplete records



Duplicates

Remove duplicate transaction records



Schema

Validate columns and data types



Formats

Standardize dates, categories and store codes



Malformed rows

Isolate invalid input records

Gold Layer — Retail Analytics



RISK

Sales performance metrics



TRENDS

Population / health trends



QUALITY

Validated analytical outputs



DASHBOARD

Consumption-ready tables

TARGET

`grocery_catalog.gold.sales_metrics`

Microsoft Azure databricks

Search data, notebooks, recent, and more... CTRL + P

grocery-sales-dashboards

Microsoft Azure

Workspace

Recent

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Query

Dashboard

Service Agents

Alerts

Query History

SQL Warehouse

Data Engineering

Runs

Data Inspection

Visual Data Prep

AI/ML

Playground

Agents

AI Gateway

Experiments

Features

Models

Serving

Grocery Sales Dashboard

Date | Untitled page | +

Global filters

Total Sales

1,073,455,279

1.07B

Monthly Sales Trend

Line chart showing monthly sales trend from 2013-01 to 2017-08. The Y-axis represents the sum of sales in millions, ranging from 10M to 30M. The trend shows significant fluctuations, with a notable peak in late 2016 and a sharp decline in early 2017.

Top 10 Stores by Sales

Bar chart showing the top 10 stores by sales. The Y-axis represents the sum of sales in millions, ranging from 0 to 100M. The X-axis lists store IDs. Store 44 has the highest sales, followed by Store 45 and Store 47.

Visualization 1

Donut chart showing the distribution of promotion status. The chart is divided into two segments: Promotion (54.70%) and No Promotion (45.30%).

Sales by Holiday Type

Bar chart showing sales by holiday type. The Y-axis represents the sum of sales in millions, ranging from 0 to 100M. The X-axis lists holiday types. The 'Holiday' category shows the highest sales, followed by 'Christmas' and 'New Year'.

Top 10 Product Families by Sales

Bar chart showing the top 10 product families by sales. The Y-axis represents the sum of sales in millions, ranging from 0 to 300M. The X-axis lists product families. 'GROCERY 1' has the highest sales, followed by 'BEVERAGES' and 'PRODUCE'.

PyTest & Data Validation



TABLES

Required tables created



COUNTS

Outputs are non-empty



SCHEMA

Columns & types validated



DUPLICATES

Duplicate detection



VALUES

Invalid values filtered

QA GATE

Only validated datasets move forward to analytics consumption.

Alerts, Monitoring & Logging



PIPELINE FAILURE

Notify the team when ETL jobs fail.



DATA ANOMALY

Flag quality or unexpected data conditions.



SLACK

Send operational notifications in real time.

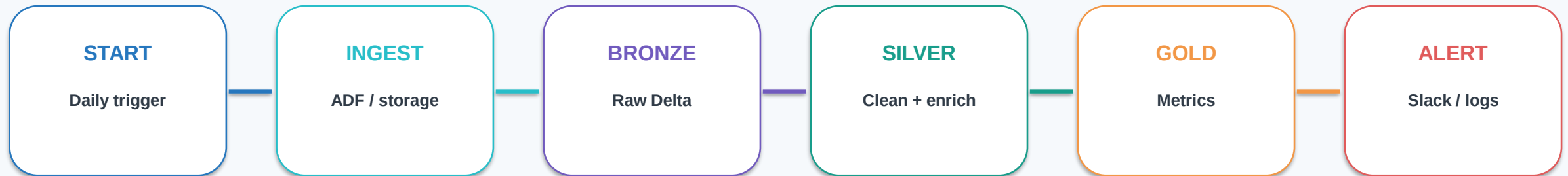


ERROR LOG

Store failures and abnormal records for troubleshooting.

`grocery_catalog.logs.etl_errors`

Airflow Pipeline Automation



ORCHESTRATION RESPONSIBILITIES

Schedule daily execution and manage dependencies.

Trigger Databricks notebooks/jobs for each medallion layer.

Monitor task status and route failures to alerting.

Provide a repeatable, automated production workflow.

Business Outcome

A scalable, automated grocery data platform that turns raw retail data into trusted business insights.



Scalable

Supports daily retail data processing.



Reliable

Improves data quality and traceability.



Actionable

Produces sales, inventory and customer metrics for analytics.



Automated

Reduces manual monitoring and operational effort.



Extensible

Ready for additional datasets and predictive use cases.

RAW → TRUSTED → ANALYTICS → DECISION SUPPORT

THANK YOU